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In [4]: import matplotlib.pyplot as plt
import numpy as np
```

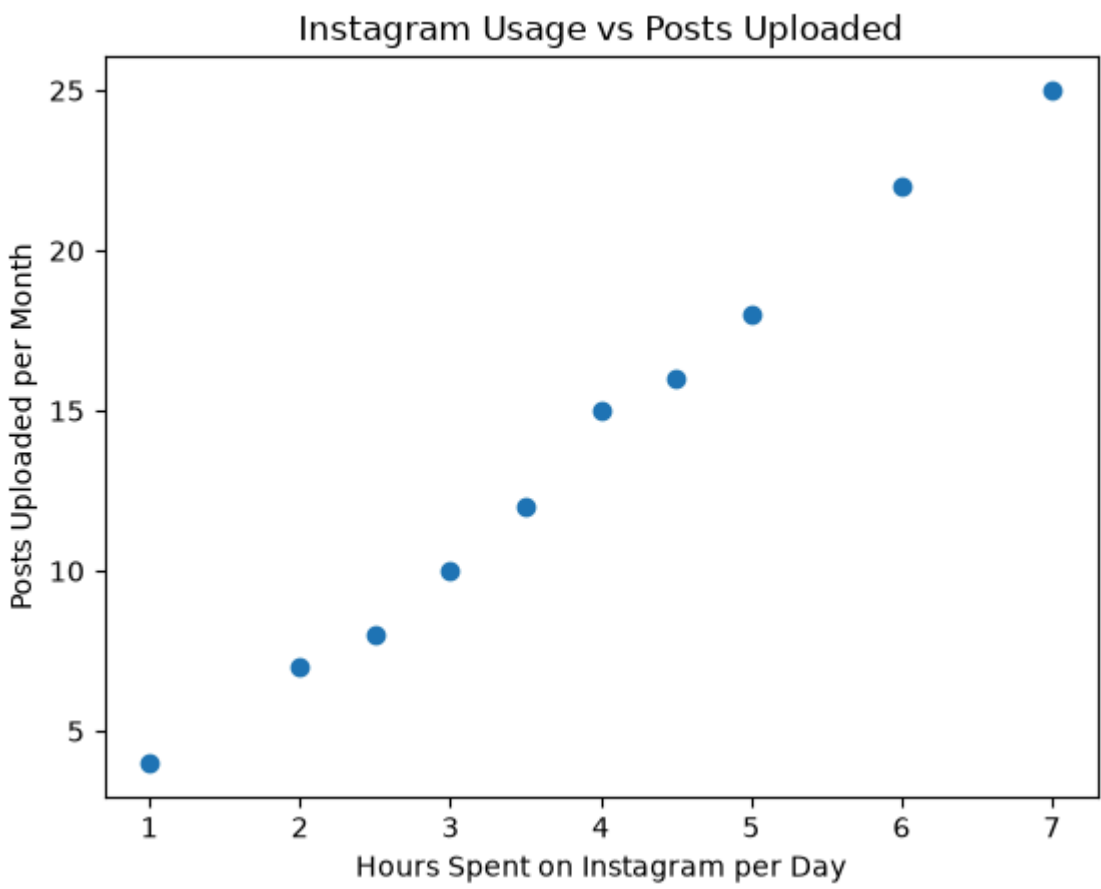
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In [5]: #Task 1 – Instagram Hours vs Posts Uploaded

instagram_hours = [1, 2, 3, 4, 2.5, 5, 3.5, 6, 4.5, 7]

posts = [4, 7, 10, 15, 8, 18, 12, 22, 16, 25]

plt.scatter(instagram_hours, posts)
plt.title("Instagram Usage vs Posts Uploaded")
plt.xlabel("Hours Spent on Instagram per Day")
plt.ylabel("Posts Uploaded per Month")

plt.show()
```



```
In [6]: #Task 2 – Zomato Spending vs Number of Orders

friends = ["Rahul", "Amit", "Neha", "Priya", "Karan", "Riya", "Jay", "Megha", "Vishal", "Pooja", "Rohan", "Ankit"]

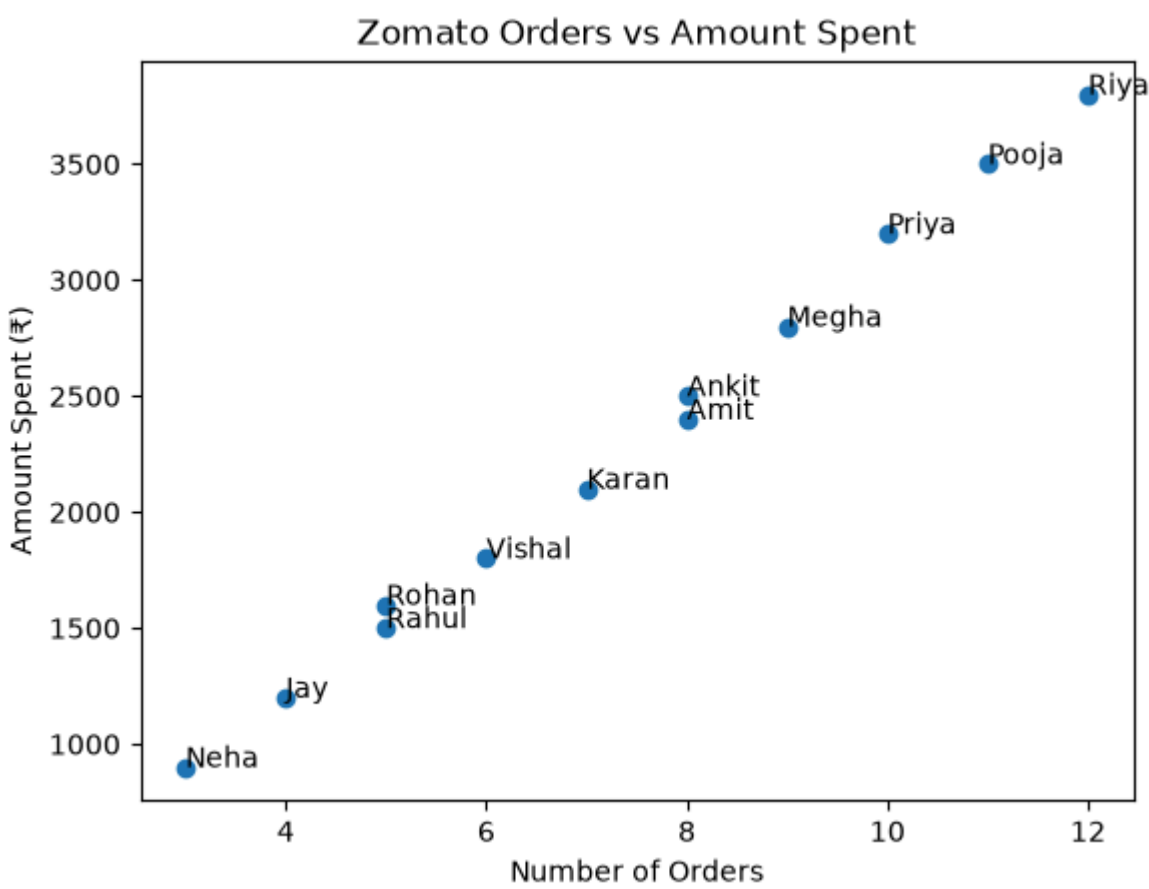
orders = [5, 8, 3, 10, 7, 12, 4, 9, 6, 11, 5, 8]

amount_spent = [1500, 2400, 900, 3200, 2100, 3800, 1200, 2800, 1800, 3500, 1600, 2500]

plt.scatter(orders, amount_spent)

for i in range(len(friends)):plt.annotate(friends[i],(orders[i], amount_spent[i]))

plt.title("Zomato Orders vs Amount Spent")
plt.xlabel("Number of Orders")
plt.ylabel("Amount Spent (₹)")
plt.show()
```

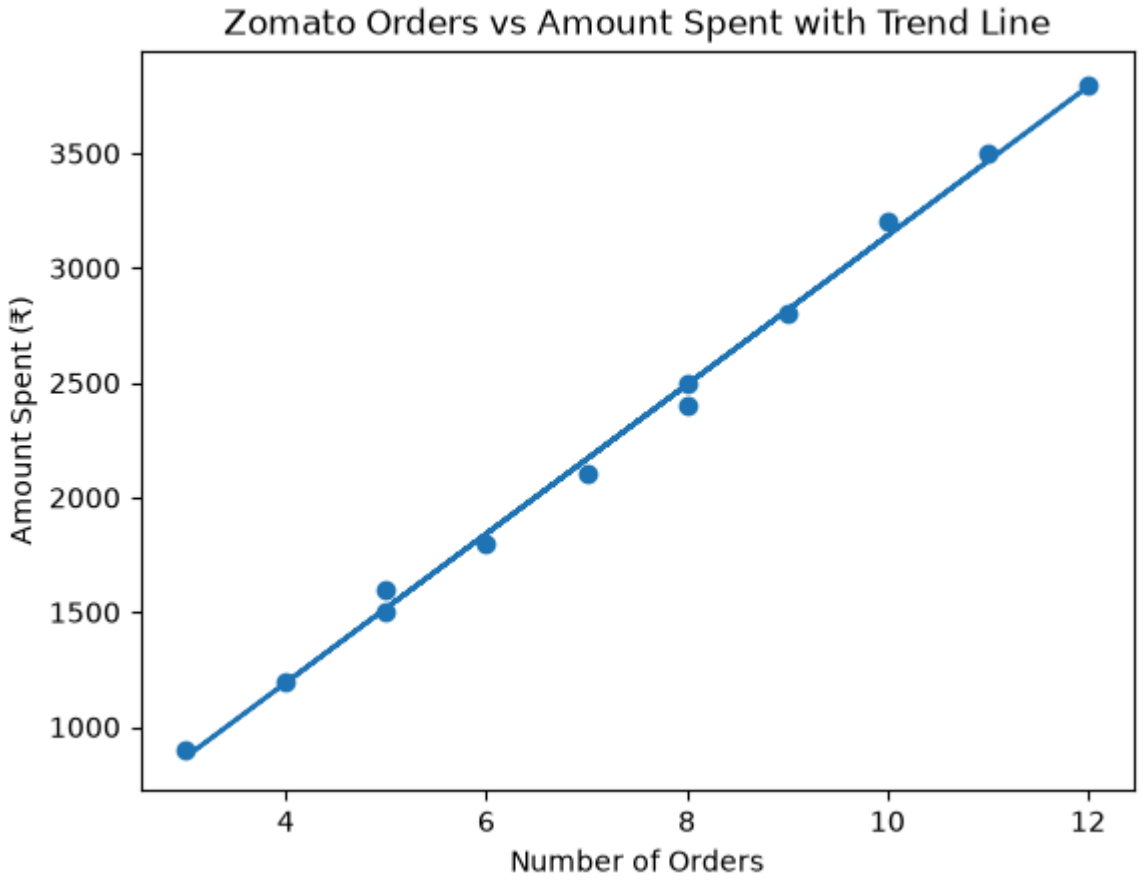


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In [7]: #Task 3 – Add Trend Line Using np.polyfit()

orders = [5, 8, 3, 10, 7, 12, 4, 9, 6, 11, 5, 8]

amount_spent = [1500, 2400, 900, 3200, 2100, 3800, 1200, 2800, 1800, 3500, 1600, 2500]

plt.scatter(orders, amount_spent)
slope, intercept = np.polyfit(orders, amount_spent, 1)
trend_line = slope * np.array(orders) + intercept
plt.plot(orders, trend_line)
plt.title("Zomato Orders vs Amount Spent with Trend Line")
plt.xlabel("Number of Orders")
plt.ylabel("Amount Spent (₹)")
plt.show()
```



```
In [8]: #Task 4 – YouTube vs Spotify Listening Hours

youtube_hours = [5, 8, 10, 12, 7, 15, 9, 11, 14, 6, 13, 16, 8, 18, 20]

spotify_hours = [3, 5, 6, 8, 4, 10, 6, 7, 9, 3, 8, 11, 5, 12, 13]

plt.scatter(youtube_hours, spotify_hours)
slope, intercept = np.polyfit(youtube_hours, spotify_hours, 1)

trend_line = (slope * np.array(youtube_hours) + intercept)

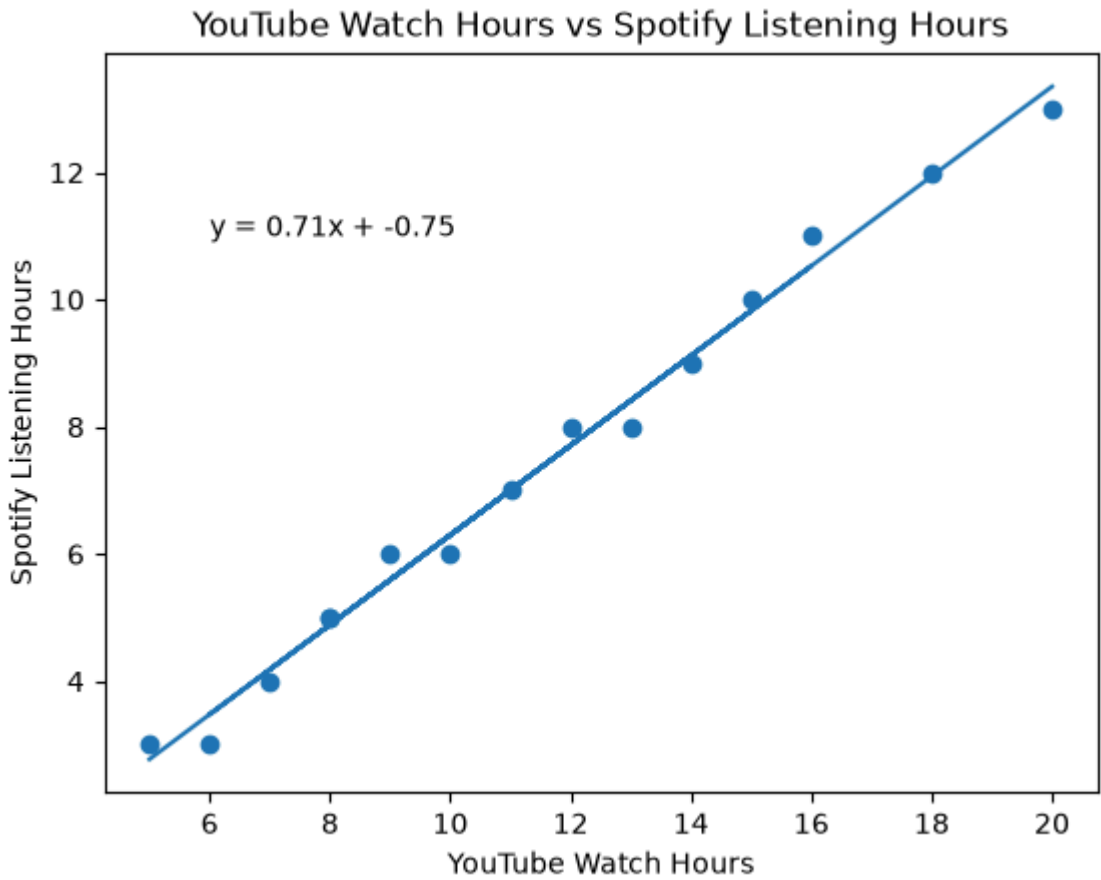
plt.plot(youtube_hours, trend_line)

equation = f"y = {slope:.2f}x + {intercept:.2f}"

plt.text(6, 11, equation)

plt.title("YouTube Watch Hours vs Spotify Listening Hours")
plt.xlabel("YouTube Watch Hours")
plt.ylabel("Spotify Listening Hours")

plt.show()
```



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In [9]: #Task 5 – Flipkart Orders vs Total Amount Spent

orders = [2, 4, 3, 6, 5, 8, 7, 10, 4, 9]
amount_spent = [1500, 3000, 2200, 4500, 3800, 6000, 5200, 7500, 3200, 6800]

plt.scatter(orders, amount_spent)

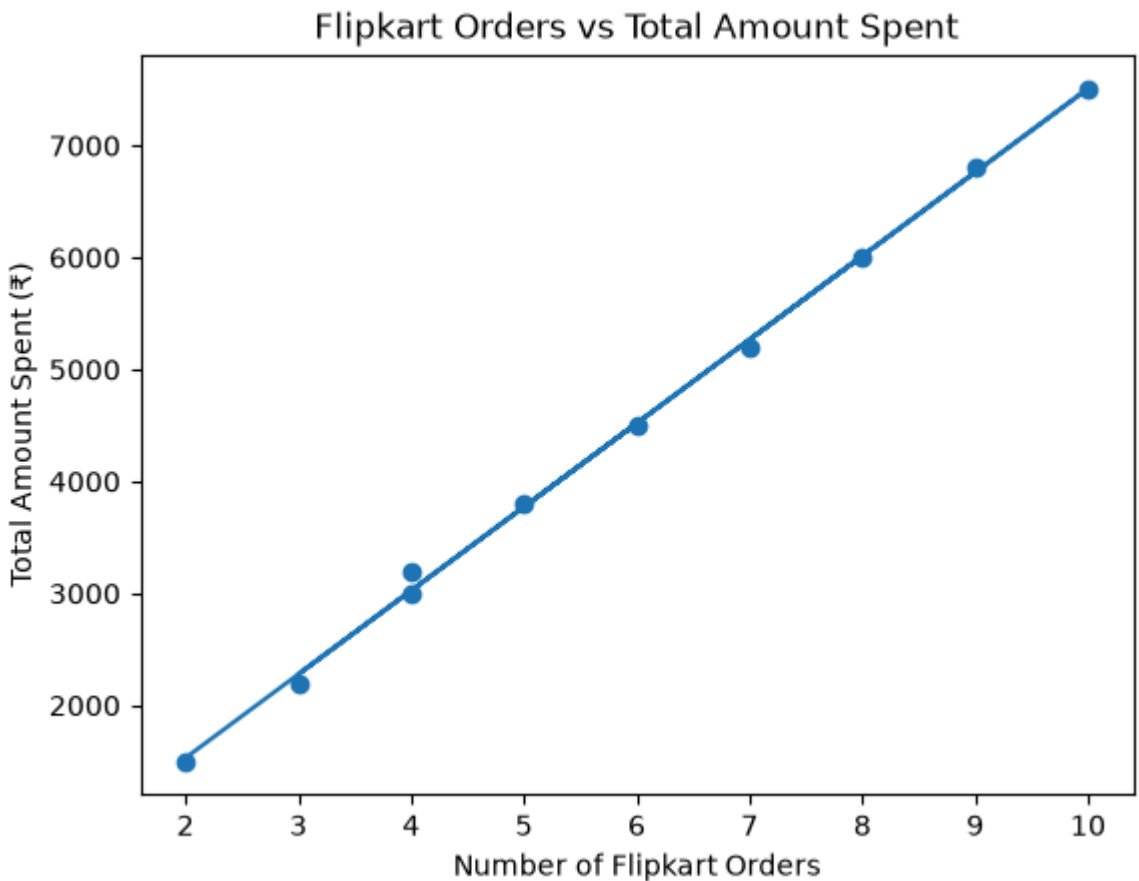
slope, intercept = np.polyfit( orders, amount_spent, 1)

trend_line = slope * np.array(orders) + intercept

plt.plot(orders,trend_line)

plt.title("Flipkart Orders vs Total Amount Spent")
plt.xlabel("Number of Flipkart Orders")
plt.ylabel("Total Amount Spent (₹)")

plt.show()
```



In []: